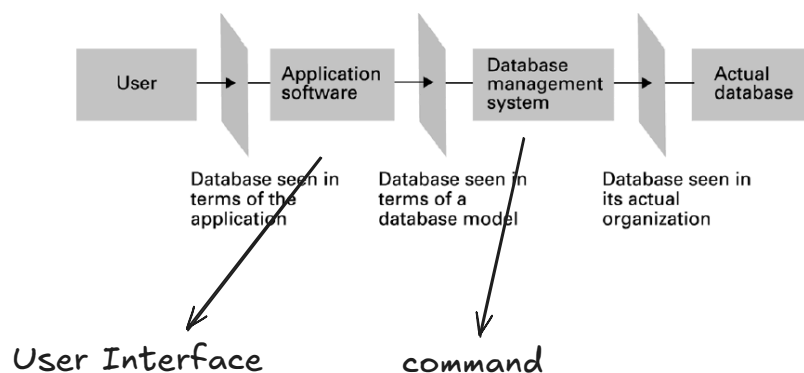


Database Fundamental

- A multidimensional(多維度) collection of data.
- Database Abstraction



- **Schemas**
 - Schema: A description of the structure of an entire database.
 - Subschema
 - A description of only that portion of the database pertinent to a **particular user's needs**.
 - Prevents sensitive data from being accessed by unauthorized personnel
- **Database Management Systems**
 - A software layer that manipulates a database in response to requests from applications
 - **Distributed Database**: The database is stored on a single machine or scattered among many machines within a **network**.
 - **Data Independence**: The ability to change the organization of a database without changing the application software that uses it.
- **Database Model**: A conceptual view of database
 - **The Relational Model**: Seen the data as **tables**.
 - Tuple(row): Information
 - Attribute(column): characteristic

Tuple

Attribute

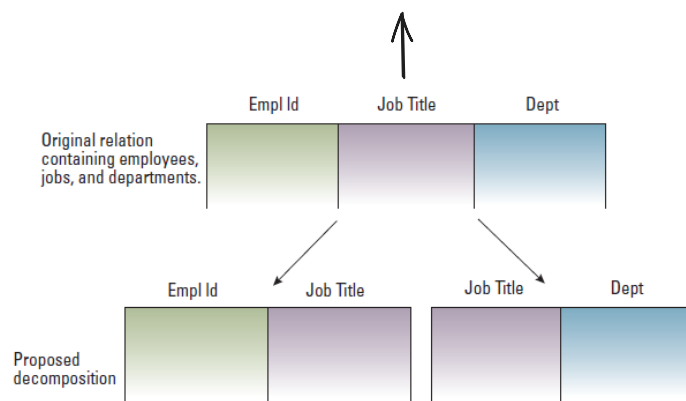
Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555
•	•	•	•
•	•	•	•
•	•	•	•

- **Obeject-oriented Database Model**

The Relational Model

- **Problems Source: Redundancy of data**
 - **Redundancy of data**: The same information appears manytimes.
 - **Deletion Anomaly(删除異常)**: A data only appear in a tuple, if delete that tuple, the **"only" data will be deleted**.
 - **Update Anomaly(更新異常)**: If you want to change value, you need to change all tuples. if you miss one, the data becomes inconsistantly.
 - **Insertione Anomaly(插入異常)**: You cannot add a new piece of data because it **depends on a primary key that doesn't exist yet**.
- **Solution**
 - **Decomposition**: Devides the tables into many parts according to their relationship.
 - Common Attribute: A main information which can **link each parts into the original table**.

Common Attribute



- **Relational Operations**

- **Select:** Choose rows
- **Project:** Choose columns
- **Join:** Merge two or many relations according to their relationship.

- **SQL (Structured Query Language)**

- **SELECT** ← Project
- **WHERE** ← Select
- **FROM**
- **INSERT**
- **UPDATE**
- **DELETE**
-